

Public-data evidence ceiling and submission audit — v5.0

Matrix-stiffness transcriptomics and osteoporosis | 9 September 2026

Editorial assessment

The project now approaches the strongest design attainable without new wet-laboratory experiments. It combines a documented systematic search, independent-study inference, REML random-effects models with modified Hartung–Knapp uncertainty, exact randomization tests, a human-only sensitivity analysis, cohort deduplication, single-cell and spatial localization, chromatin accessibility, genetics and direct bone-mechanics outcomes. Its publishable advance is a quantified boundary: matrix-stiffness responses are distributed, lineage-localized and context-dependent, while a universal gene signature and osteoporosis biomarker are unsupported.

Communications Biology remains an appropriate stretch target because its scope explicitly includes secondary data analysis and evaluates novelty, technical soundness, evidential strength and influence on thinking in the field. Editorial risk is driven by the negative/boundary-centered result rather than a remaining statistical weakness.

Evidence architecture

Layer	Data and inference	Result	Supported claim
Discovery	7 stiffness studies; 13 contrasts; within-series aggregation	0/12,241 genes at FDR; study rho -0.253 to 0.136	No demonstrable universal directional program
Pathways	1,008 sets; one score per study	109 pooled-gene signals reduced to 0 at the study unit	Correlated genes can create false precision
Lineage	Human femur single-cell, spatial and independent GSE147390 programs	Spatial program: 6/7 studies, exact sign-flip $P=0.0156$; independent program partial	Response magnitude localizes to early osteoblast lineage
Disease	3 independent human cohorts; complete label spaces; equal-cohort synthesis	Combined $P=0.259$ magnitude, 0.254 signed, 0.095 signature	Disease transfer is context-dependent
Species	Repeat meta-analysis after excluding 2 rat studies	0/12,757 genes at FDR; rho=0.810 with full effects	Main conclusion is not driven by rat data
Orthogonal challenges	ITGA11 states, ATAC, eBMD GWAS, mouse bone mechanics	Fixed signature unsupported	No biomarker or mechanical-property claim

Independent osteoporosis cohorts

Cohort	Design	Absolute rho	Exact P	Signature P	Interpretation
GSE156508	6 osteoporotic fracture vs 6 severe OA; primary osteoblasts	0.02541	0.0303	0.2229	Weak magnitude association only
GSE35958	5 primary OP vs 4 age-matched controls; cultured BM-MSCs	0.00094	0.5000	0.0238	Isolated signature result; no replication
GSE230665	12 postmenopausal OP vs 3 controls; femur tissue	-0.01097	0.7033	0.6308	No support; only 3 controls
Equal-cohort synthesis	Each cohort standardized to its complete exact null	0.3848*	0.2589	0.0950	*Combined standardized statistic; null overall

Independence rules were frozen before outcome inspection. GSE35956 was excluded because it reused the GSE35958 osteoporosis cases and used younger controls. GSE157322 was excluded because BMD group was fully confounded with ancestry. These decisions prevent inflation of cohort count by duplicate participants or structural confounding.

Human-only sensitivity

After removing rat GSE181512 and GSE310514, the same random-effects procedure was applied to five human stiffness studies. Among 12,757 genes represented in at least three studies, none passed FDR correction. Human-only and seven-study pooled effects correlated at Spearman rho=0.810. Repeating disease validation with the human-only model produced combined $P=0.199$, 0.138 and 0.0835 for magnitude, direction and signature endpoints, respectively.

Claim boundary

Suitable for title/abstract	Exploratory only	Unsupported
Stiffness-response magnitude localizes to osteoblast-lineage programs	Small GSE156508 magnitude association	A stiffness signature diagnoses osteoporosis
Directional effects show poor cross-study agreement	Single-cohort GSE35958 signature difference	A stable core gene set was discovered
Human-only analysis preserves the main conclusion	Residual spatial signal after overlap removal	Transcriptomics measures bone material stiffness
Combined disease tests do not support universal transfer	One-donor GSE147390 partial support	Causality, fracture prediction or treatment response

Resolved submission metadata

- Author order, affiliations, equal-contribution statement, corresponding-author contact and ORCID supplied.
- Funding, CRediT contributions, competing-interest statement and public-data ethics statement supplied.
- Archived release DOI 10.5281/zenodo.22726636 retained in the manuscript and submission metadata.
- Retain the disclosure that the revision-stage search was not prospectively registered.
- Do not convert the heterogeneous disease cohorts into a clinical-prediction claim.

Journal sequence

Priority	Journal	Positioning	Risk
1	Communications Biology	Mechanistic boundary, reproducibility and rigorous secondary synthesis	Medium-high: editor must value a negative result that changes interpretation
2	iScience	Public multi-omics integration and reproducibility	Medium
3	JBMR Plus / Bone Reports	Bone-biology translational boundary	Lower; narrower reach
Not first choice	Nature Communications / Science Advances	Usually require stronger mechanism or prospective validation	Very high without new experiments

Additional routine machine learning, WGCNA or network pharmacology would not raise the evidential ceiling and could reduce credibility. The only clear non-experimental route to a materially higher ceiling is a newly available, larger human bone single-cell/spatial cohort with complete donor metadata, or a clinical collaboration providing external BMD/fracture outcomes.

Target-journal format check

Requirement	Status
Article title up to 15 words	Pass: 14 words
Abstract below 150 words	Pass: 137 words
Statistics and reproducibility section	Pass
Data and Code availability	Present; archived release DOI supplied
PRISMA checklist	Complete; non-registration disclosed
Figures in text plus high-resolution files	Pass: PNG/PDF/TIFF
AI-use disclosure in Methods	Present; authors accept responsibility

Primary sources

Communications Biology, Aims & Scope: <https://www.nature.com/commsbio/aims>

Communications Biology, Submission guidelines: <https://www.nature.com/commsbio/submit/submission-guidelines>

Communications Biology, Content types: <https://www.nature.com/commsbio/submit/content-types>

NCBI GEO GSE156508: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE156508>

Benisch et al., PLoS ONE 2012 (GSE35958): <https://pmc.ncbi.nlm.nih.gov/articles/PMC3454401/>

NCBI GEO GSE230665: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE230665>

PRISMA 2020 checklist: <https://www.prisma-statement.org/prisma-2020-checklist>

All numerical statements are generated from frozen scripts and machine-readable outputs in the submission archive.